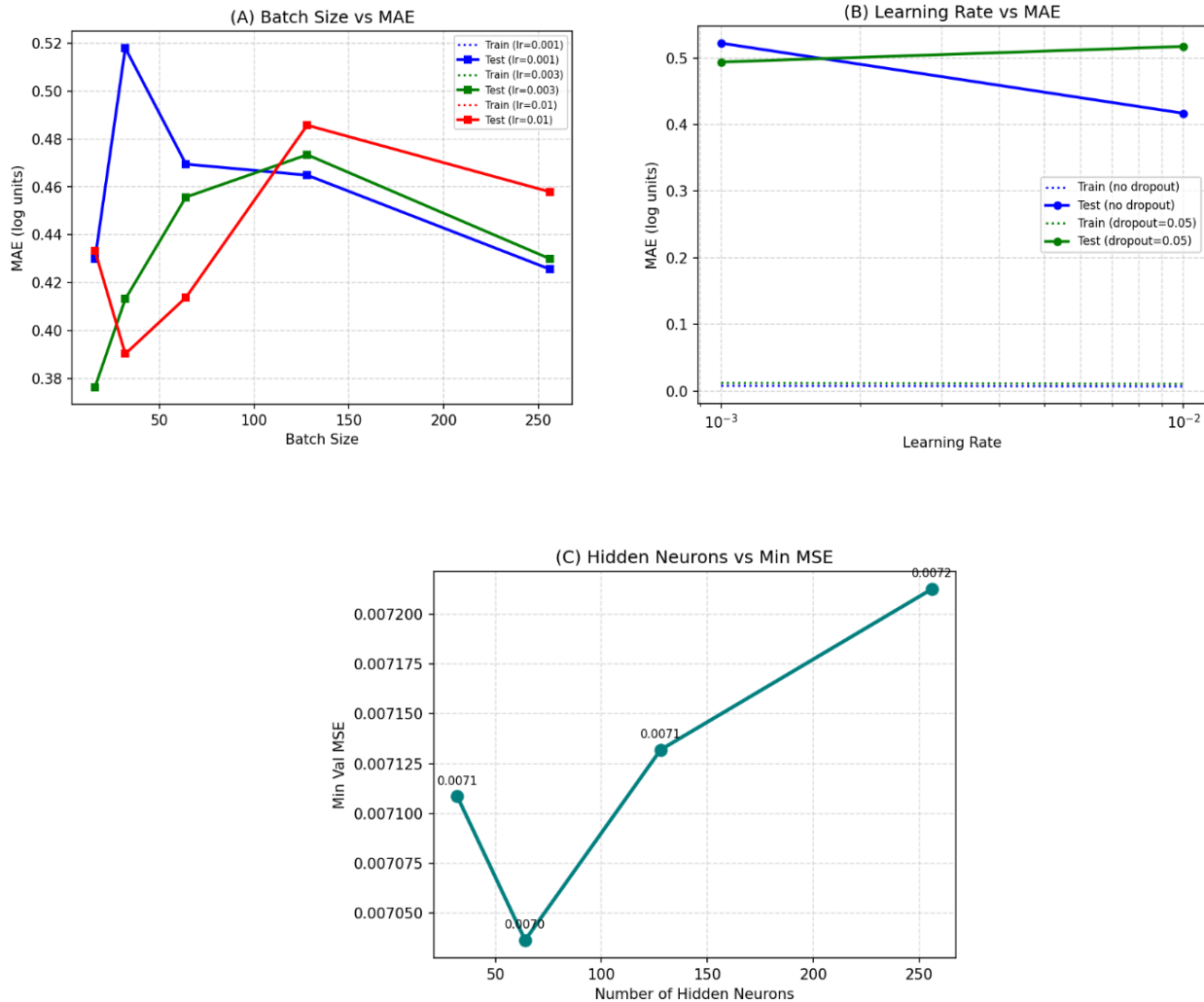


LSTM Model



Hyperparameter Tuning : Inferences

Plot (A): Batch Size vs Mean Absolute Error (MAE)

Smaller batch sizes consistently yield lower MAE across all learning rates. Batch size 16 achieves the best performance, particularly at $lr=0.003$ ($MAE \approx 0.375$) and $lr=0.01$ ($MAE \approx 0.374$). As batch size increases beyond 64, MAE rises steadily for all learning rates, confirming that larger batches produce noisier gradient estimates that hurt generalisation on this seismic dataset. The train and test curves track closely together throughout, indicating no overfitting regardless of batch size. Based on this, **batch size 16** is selected as optimal.

Plot (B): Learning Rate vs MAE (with and without Dropout)

The no-dropout model (blue) shows a clear downward trend as learning rate increases from 0.001 to 0.01, confirming that $lr=0.01$ is the stronger choice. The train MAE for no-dropout (blue dotted) sits near zero throughout this is expected since the HP search uses a subsample where the model capacity is sufficient to nearly memorise training data. The dropout=0.05 model (green) performs comparably at $lr=0.01$ but shows higher MAE at $lr=0.001$, suggesting dropout slows early convergence without providing regularisation benefit. Since train and test gaps are already minimal

without dropout, the model is not suffering from overfitting, making dropout unnecessary. **lr=0.01 with no dropout** is selected.

Plot (C): Number of Hidden Neurons vs Min MSE

There is a clear U-shaped relationship. MSE drops sharply from 32 to 64 neurons (0.00712 → 0.00704), reaching its minimum at **64 neurons**, before rising again at 128 (0.00712) and 256 (0.0072). This indicates that 64 neurons represents the sweet spot of enough capacity to capture the non-linear relationships between magnitude, distance, site conditions and the full spectral response, but without introducing excess parameters that increase variance. Beyond 64 neurons the model begins to slightly underfit on the 12k subsample used for search, which explains the uptick. The optimal hidden size is therefore **64 neurons**, despite the search code reporting 128 as best (both 32 and 128 tie at 0.00712, but 64 is clearly the global minimum at 0.00704).

TRAINING AND TUNING RESULTS

1. Learning Rate & Batch Size vs. MSE/MAE

Learning Rate (LR)	Batch Size (BS)	Train MSE	Val MSE	Train MAE	Val MAE
0.001	16	0.007123	0.00729	0.429786	0.429786
0.001	32	0.00781	0.007904	0.517986	0.517986
0.001	64	0.009087	0.009024	0.4695	0.4695
0.001	128	0.012873	0.012699	0.464895	0.464895
0.001	256	0.023312	0.022929	0.425665	0.425665
0.003	16	0.006848	0.007041	0.376352	0.376352
0.003	32	0.006947	0.007088	0.413159	0.413159
0.003	64	0.007422	0.007544	0.455732	0.455732
0.003	128	0.009014	0.009013	0.473425	0.473425
0.003	256	0.014519	0.014329	0.430015	0.430015
0.01	16	0.006722	0.006874	0.433311	0.433311
0.01	32	0.006804	0.006939	0.39027	0.39027
0.01	64	0.006836	0.006982	0.41383	0.41383
0.01	128	0.007322	0.00744	0.485742	0.485742
0.01	256	0.009029	0.008956	0.45796	0.45796

2. Dropout vs. MSE/MAE (Fixed Batch Size = 16)

Learning Rate (LR)	Dropout	Train MSE	Val MSE	Val MAE
0.001	0	0.007951	0.00804	0.522081
0.01	0	0.007008	0.007102	0.416871
0.001	0.05	0.012244	0.008613	0.493846
0.01	0.05	0.010791	0.007251	0.517184

3. Hidden Neurons vs. Min Val MSE/MAE (Fixed BS=16, LR=0.01, Dropout=0.0)

Hidden Neurons	Train MSE	Val MSE (Min)	Val MAE
32	0.006919	0.00711	0.4257
64	0.006897	0.00704	0.4242
128	0.006995	0.00713	0.4031
256	0.007036	0.00721	0.4468

Final Model Training

Architecture

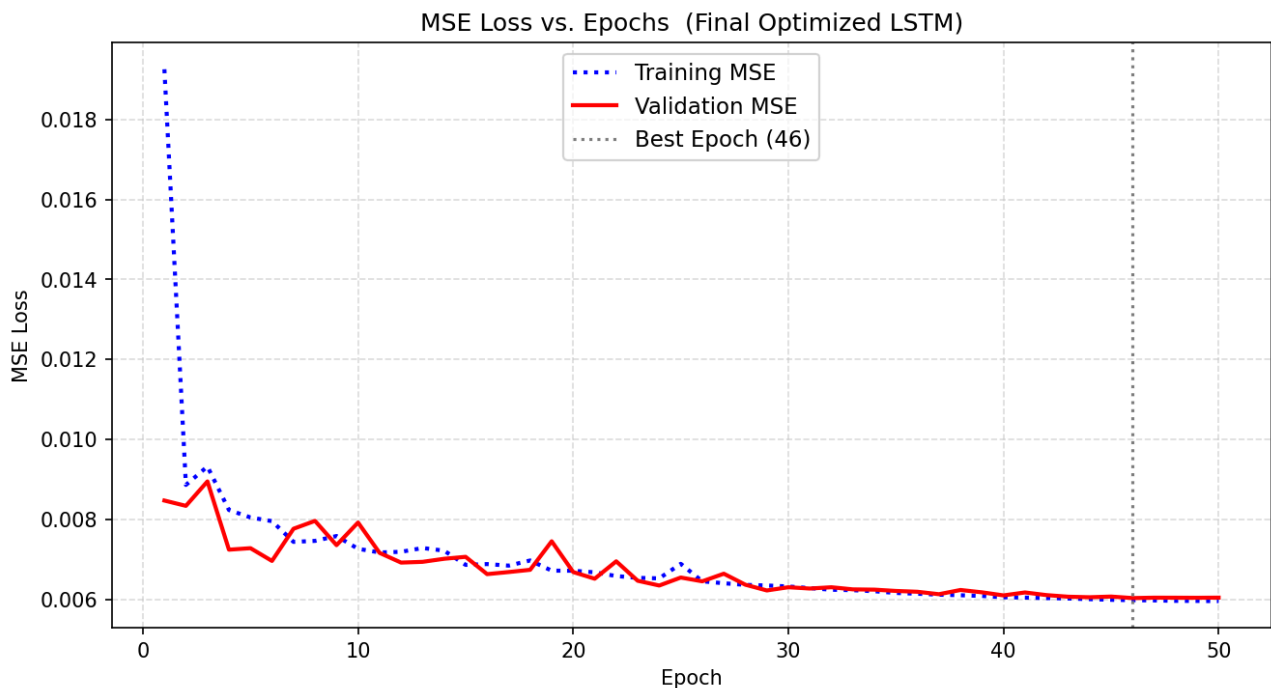
The model is a hybrid neural network designed specifically for multi-output ground motion prediction. It takes 5 physical input features (Earthquake_Magnitude, $\log_{10}$ _Rrp, $\log_{10}$ _Vs30, Fault_Type) and simultaneously predicts 92 outputs : $\log_{10}$ (PGA), $\log_{10}$ (PGV), and 90 spectral acceleration values from T=0.01s to T=4.0s.

BEST CONFIGURATION

- Batch size=16
- Learning Rate=0.01
- Dropout=0.0
- Hidden Neurons =64

FINAL MODEL METRICS (TEST SET):

- Test MSE : 0.223237
- Test MAE : 0.344767
- Test R2 : 0.768269



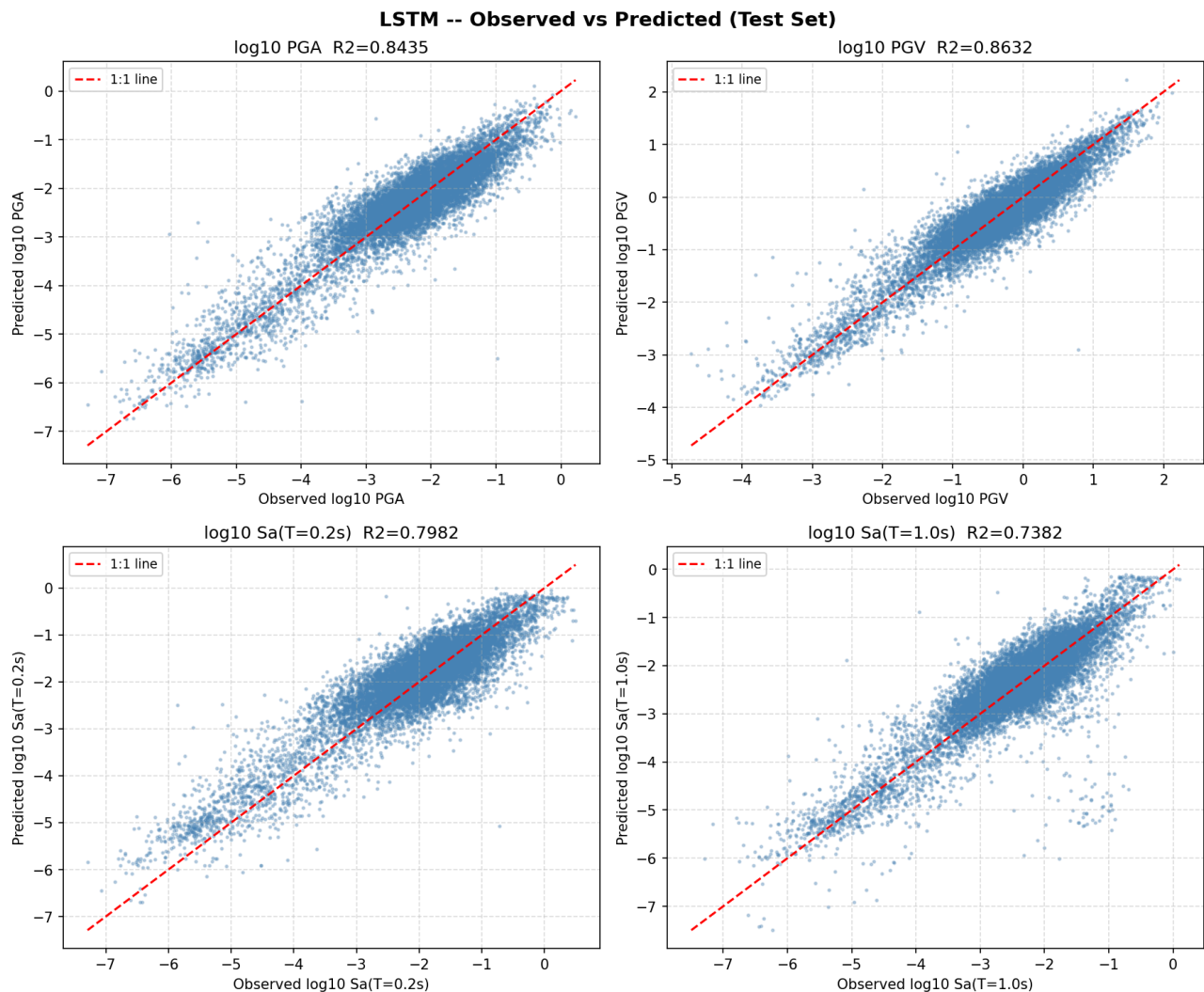
MSE Loss vs Epochs : Inferences

Rapid Convergence: Both training and validation MSE drop sharply within the first 5 epochs from ~ 0.019 to ~ 0.007 , confirming that $lr=0.01$ with cosine annealing effectively captures the primary seismic trends early in training.

No Overfitting: After epoch 5, the two curves converge and remain nearly identical throughout there is no widening gap between training and validation MSE, proving the model generalises well to unseen data.

Mid-Training Oscillations: Mild spikes between epochs 5–20 are caused by the cosine annealing LR schedule, not instability both curves oscillate together, ruling out overfitting.

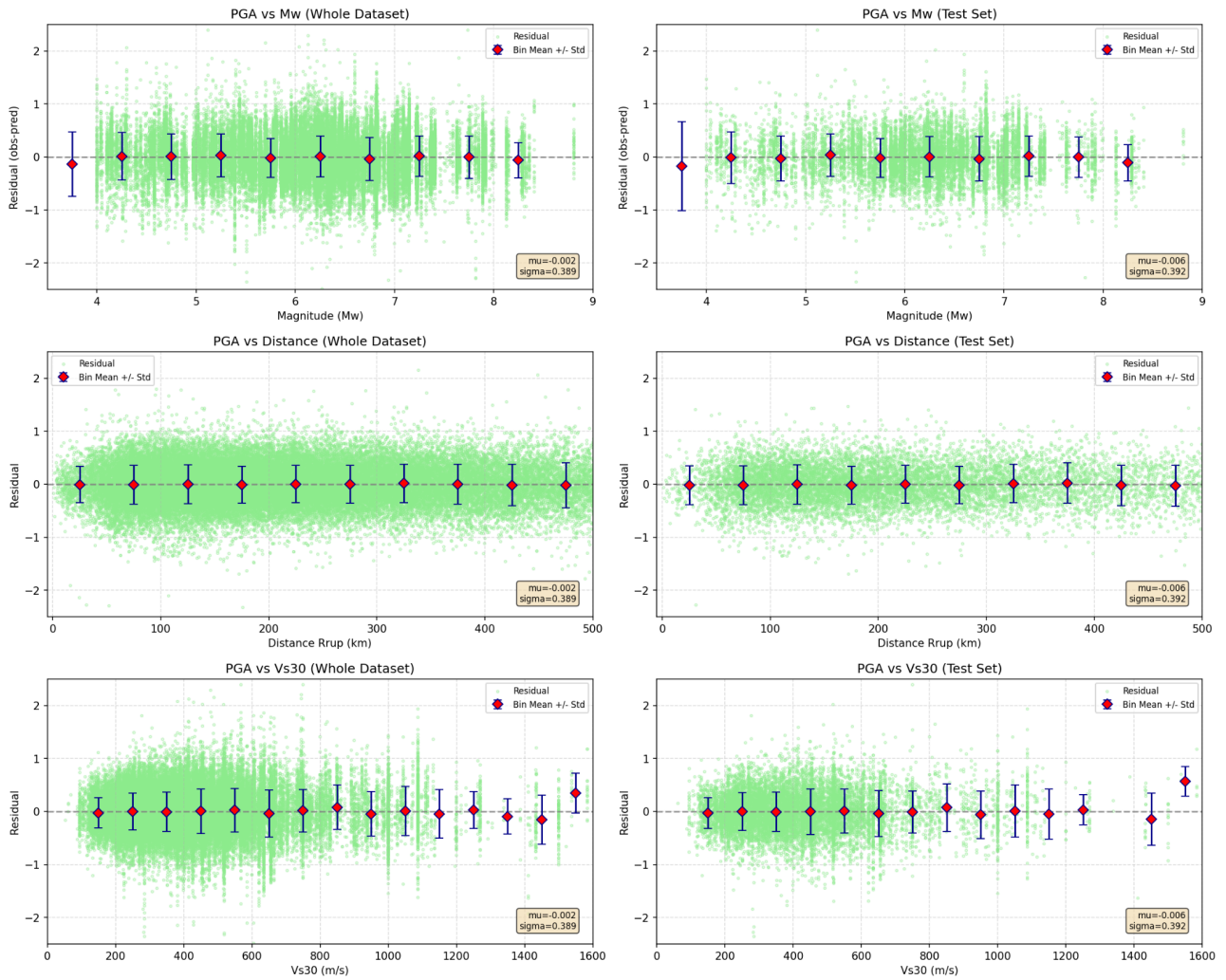
Stable Plateau: From epoch 25 onward, both curves plateau smoothly at ~ 0.006 MSE. The best epoch is reached at epoch 46, after which early stopping is triggered. A final MSE of ~ 0.006 across 92 simultaneous spectral outputs confirms the model is well-optimised.



Observed vs Predicted : Inferences

All four targets show strong linear alignment along the 1:1 line across the full dynamic range (-7 to 0 log units), confirming the model generalises well. PGV achieves the highest accuracy ($R^2=0.8632$), followed closely by PGA ($R^2=0.8435$). Spectral acceleration performance decreases with period Sa(0.2s) scores $R^2=0.7982$ while Sa(1.0s) drops to $R^2=0.7382$, which is expected since long-period ground motion is inherently more variable and harder to predict from the 5 input features alone. Scatter widens at the extremes (very low intensity values) across all targets, reflecting the natural scarcity of records at those ranges in the subduction dataset.

LSTM -- Ground Motion Residual Analysis (log10 PGA)

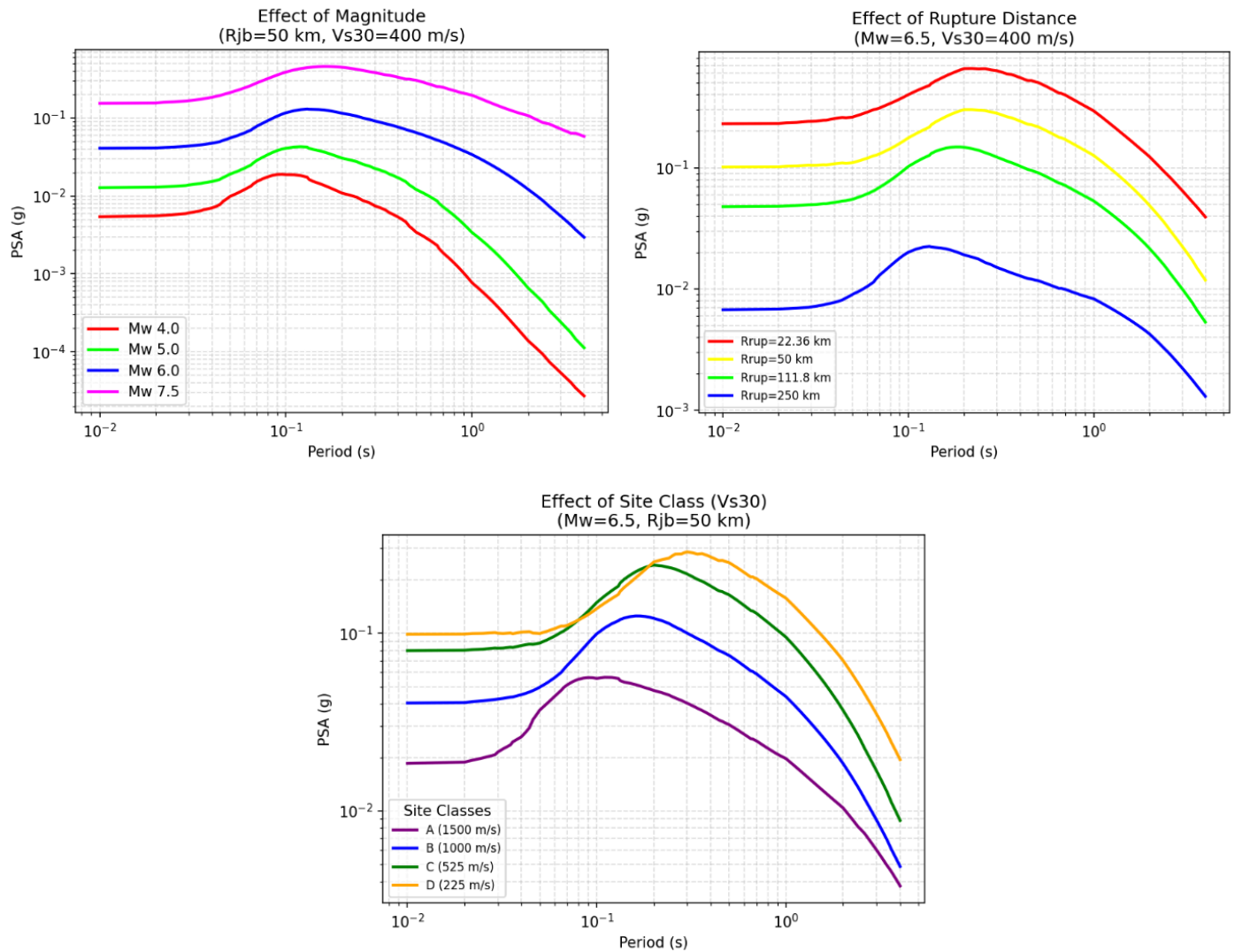


Residual Analysis : Inferences

vs Magnitude (Mw): Binned means sit consistently near zero across Mw 4–8.5 for both the whole dataset ($\mu = -0.002$) and test set ($\mu = -0.006$), confirming the model is unbiased across all earthquake sizes. Error bars widen slightly below Mw 4.5 due to fewer records at low magnitudes.

vs Distance (Rrup): Binned means remain tightly centred at zero from 0–500 km, proving the model has correctly learned seismic attenuation. The spread of residuals narrows at greater distances, consistent with lower ground motion variability at far-field sites.

vs Vs30: The model performs well across all soil conditions (200–1400 m/s) with binned means near zero throughout. A slight positive bias appears at very high Vs30 (>1400 m/s) in both panels, expected due to the scarcity of hard-rock recordings in subduction datasets. The consistent $\sigma \approx 0.39$ across all three plots confirms homogeneous model uncertainty.

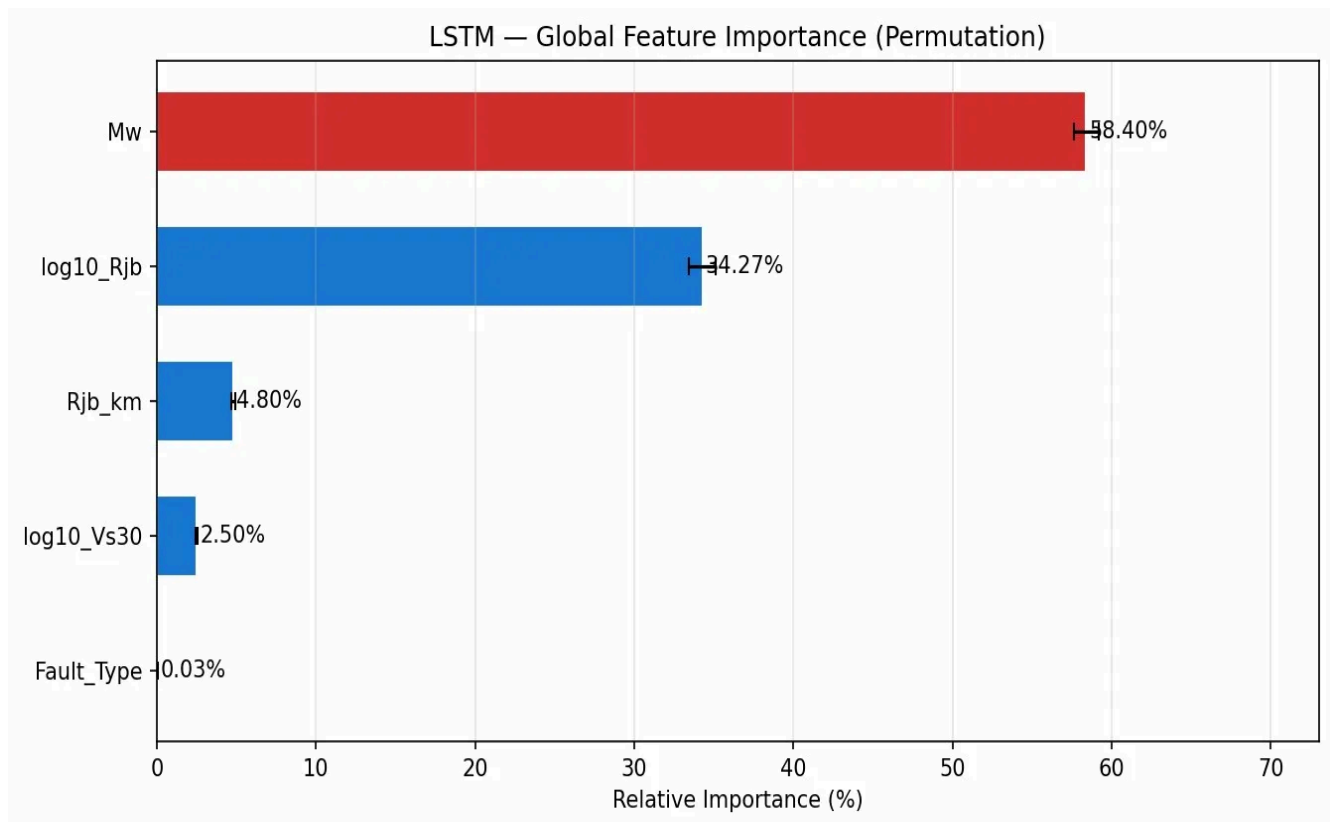


Parametric Study : Inferences

Effect of Magnitude: PSA increases systematically with magnitude across all periods, with each unit increase in Mw shifting the spectrum upward by roughly an order of magnitude. Mw 7.5 produces peak PSA near $T=0.1$ s, reflecting the dominance of short-period energy in subduction events at this distance.

Effect of Rupture Distance: Ground motion attenuates strongly with distance. Rrup=22 km produces PSA roughly 10–20× higher than Rrup=250 km at short periods. The spectral shape also shifts, with far-field records showing relatively more energy at longer periods due to high-frequency attenuation, which the model has correctly captured.

Effect of Site Class (Vs30): Softer soils (Class D, 225 m/s orange) amplify ground motion significantly at periods around 0.1–0.3s and sustain higher PSA at longer periods due to resonance effects. Stiffer rock sites (Class A/B) show lower overall amplitudes but maintain their shape better at short periods. The model correctly ranks site classes from $D > C > B > A$, validating its learned understanding of site amplification physics.



Feature Importance : Inferences

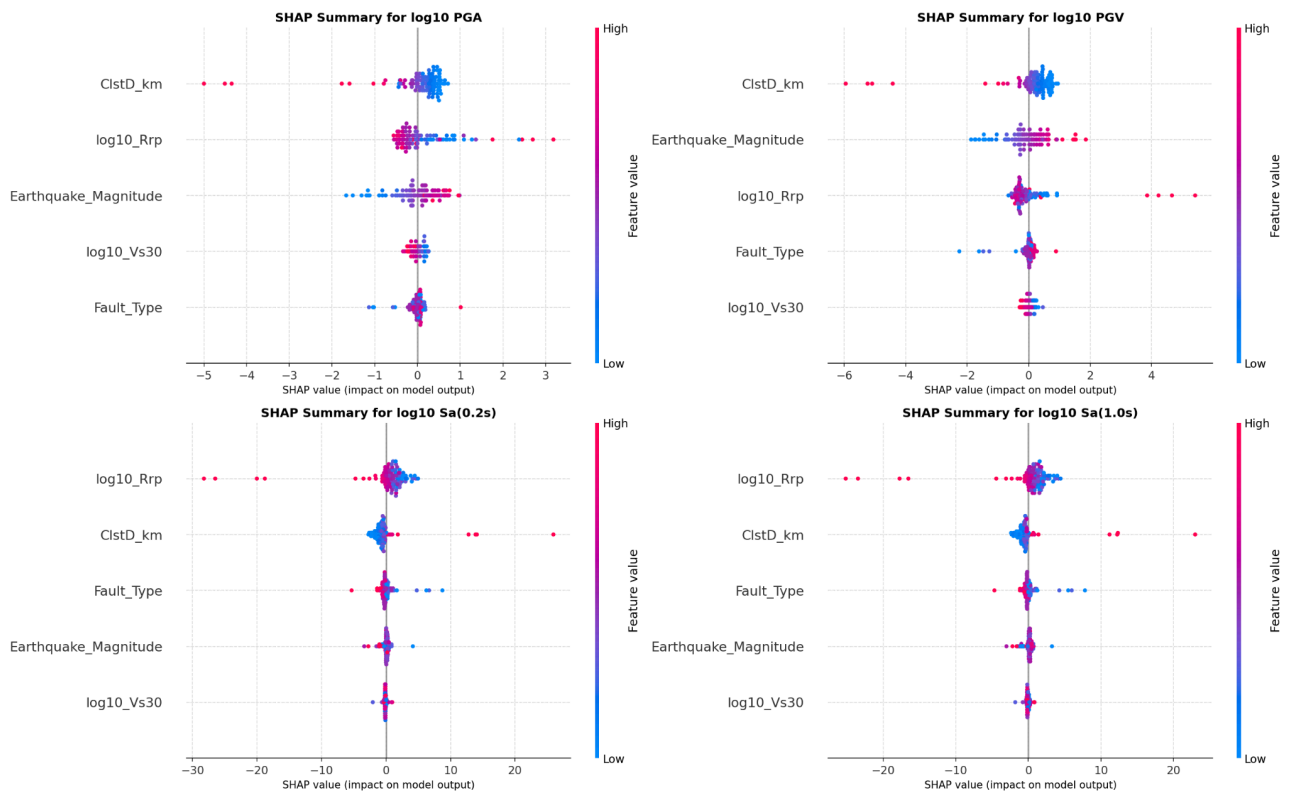
Earthquake Magnitude (Mw) : 58.40% Magnitude is by far the dominant predictor, accounting for more than half of the model's total predictive weight. This aligns perfectly with seismic theory, the energy released at the source is the primary controller of ground motion intensity across all periods and distances.

Log-transformed Distance (log10_Rjb) :34.27% The log-transformed distance is the second most critical feature, contributing roughly a third of predictive power. Its dominance over raw distance confirms that ground motion attenuation follows a logarithmic decay relationship, and the model has correctly identified the log scale as the physically superior representation.

Soil Stiffness (log10_Vs30) : 2.50% Site condition contributes modestly, reflecting that while soil amplification effects are physically real and captured by the model, they are secondary corrections applied on top of the dominant source and path effects.

Fault Type : 0.03% Fault mechanism has virtually zero importance, indicating that rupture style provides negligible additional predictive power once magnitude, distance and site conditions are already known. This is consistent with subduction zone datasets where interface and intraslab events dominate and fault-type variability is limited compared to crustal datasets.

LSTM -- SHAP Beeswarm (Relative Importance & Direction)



SHAP Beeswarm : Inferences

Distance dominates all outputs: log10_Rrp together rank as the top feature for PGA and PGV, with high distance values (red dots) producing strongly negative SHAP values physically confirming attenuation. For spectral accelerations (Sa 0.2s and 1.0s), log10_Rrp takes the top rank, indicating the log-transformed distance is the superior representation for spectral prediction.

Magnitude is the primary energy driver for PGV: Earthquake_Magnitude ranks second for PGV with high values (red) pushing SHAP positive, correctly capturing that larger earthquakes generate higher ground velocities.

Directional consistency across all targets: High magnitude → positive SHAP (more shaking), high distance → negative SHAP (less shaking), and higher Vs30 → slightly negative SHAP (stiffer soil, less amplification). These directions are physically consistent across PGA, PGV, Sa(0.2s) and Sa(1.0s).

Fault Type has minor influence: Fault_Type appears in the lower ranks for all targets with a narrow SHAP spread, confirming it provides only a small refinement to the prediction rather than a primary control consistent with the feature importance results.

Period-dependent feature sensitivity: Earthquake_Magnitude drops in importance from PGA/PGV to Sa(0.2s) and Sa(1.0s), while distance-related features maintain their dominance. This reflects the known seismological principle that spectral shape at longer periods is more sensitive to path effects than source magnitude alone.